Product Modification, Legal Pricing Constraints, and Third-Degree Price Discrimination

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Abstract

We study how a monopolist manufacturer circumvents a common-wholesale-price constraint by introducing low-cost product variants. In a two-retailer linear-demand model, a legal constraint (modeled on the Robinson–Patman Act) forces a single wholesale price for identical goods. The manufacturer can pay a per-unit cost $k \geq 0$ to create differentiated variants, restoring segment-specific pricing. We derive closed-form profit under both regimes, prove that variant-based discrimination is strictly profitable for generic parameter values when $k = 0$, characterize the threshold cost $\bar{k}$ as a function of demand heterogeneity and expected legal penalty L, and verify the standard Ramsey inverse-elasticity condition at the discriminatory optimum. Welfare analysis shows the output condition of Schmalensee (1981) extends to this setting, with an additional social cost $k \sum_i Q_i^D$ when $k > 0$. We provide descriptive evidence using 3.24 M scanner observations from Dominick's Finer Foods: OLS of per-unit price on ln(Size) yields $\hat{\beta} = -3.00$ (SE $= 0.37$); store fixed effects are jointly significant ($F = 15.04$); within-brand regressions for Tide, All, and Purex confirm the gradient persists within product lines.

Keywords: price discrimination, nonlinear pricing, legal constraints, product differentiation, scanner data

MSC 2020: 91B24, 91B26 *JEL:* D42, L11, L42, K21

1 Introduction

Third-degree price discrimination requires that a firm charge different prices to distinct market segments. The classical treatment [1, 2] assumes the firm can freely partition markets and set segment-specific prices. In many jurisdictions, however, wholesale price discrimination is legally constrained. In the United States, the Robinson–Patman Act (RPA, 1936) prohibits charging competing buyers different prices for commodities of "like grade and quality" [11].

Manufacturers respond by introducing *product variants*—differences in package size, labeling, or minor formulation—that place goods outside the "like grade and quality" category, effectively relaxing the uniform-price constraint. The Seventh Circuit's decision in *Woodman's Food Market v. Clorox* [12] illustrates the legal significance of this boundary. Despite its practical importance, the product-modification channel has received limited formal treatment. Yonezawa, Gómez, and Richards [6] document wholesale-price divergence across retailers but do not model the mechanism. Kobayashi and Muris [7] analyze the economics of Robinson–Patman from a policy perspective. Ross [8] studies the Act's distributional effects.

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Contributions. We make three contributions.

(i) A tractable model in which product modification is an explicit decision variable interacting with a common-price legal constraint (Section 2).

(ii) Closed-form expressions for the profit gain, threshold differentiation cost $\bar{k}$, and comparative statics in the legal penalty L (Theorems 2 and 4).

(iii) Descriptive evidence from 3.24 M scanner observations linking the model's predictions to observed size–price gradients and within-brand pricing patterns (Section 4).

2 Model

2.1 Environment

A monopolist manufacturer with constant marginal cost $c > 0$ supplies two retailers $i \in \{1, 2\}$. Retailer i faces linear demand

$$Q_i(p_i) = a_i - b_i\, p_i, \qquad a_i,\, b_i > 0, \tag{1}$$

and applies a fixed markup $m_i \geq 0$, so $p_i = w_i + m_i$. Without loss of generality, $b_1 < b_2$ (retailer 1 has less elastic demand).

Assumption 1 (Legal constraint)**.** When the manufacturer sells identical commodities to both retailers, it must charge a common wholesale price $w_1 = w_2 = w$. If it introduces distinct variants at per-unit cost $k \geq 0$, it may set $w_1 \neq w_2$.

The timing is: (i) the manufacturer chooses a pricing/product-design strategy; (ii) retailers observe wholesale prices and set retail prices $p_i = w_i + m_i$.

2.2 Uniform-pricing regime

Under the legal constraint, the manufacturer solves

$$\max_{w}\ \Pi^U(w) = (w - c)\sum_{i=1}^{2}\big(a_i - b_i(w + m_i)\big). \tag{2}$$

This is a concave quadratic in w, with unique maximizer

$$w^* = \frac{\sum_i (a_i - b_i m_i) + c\sum_i b_i}{2\sum_i b_i} \tag{3}$$

and optimal profit

$$\Pi^U = \frac{\big(\sum_i (a_i - b_i m_i) - c\sum_i b_i\big)^2}{4\sum_i b_i}. \tag{4}$$

2.3 Discriminatory regime

With distinct variants, the manufacturer independently maximizes over each w_i:

$$\Pi^D(w_1, w_2;\, k) = \sum_{i=1}^{2}(w_i - c - k)\big(a_i - b_i(w_i + m_i)\big). \tag{5}$$

The first-order conditions yield

$$w_i^* = \frac{a_i - b_i m_i + b_i(c + k)}{2b_i} = \frac{a_i}{2b_i} - \frac{m_i}{2} + \frac{c + k}{2}, \tag{6}$$

with optimal profit

$$\Pi^D(k) = \sum_{i=1}^{2}\frac{\big(a_i - b_i m_i - b_i(c + k)\big)^2}{4b_i}. \tag{7}$$

We maintain $k < \min_i \hat{k}_i$, where $\hat{k}_i := a_i/b_i - m_i - c$, so that both markets are served.

2.4 Main results

Proposition 2 (Strict profitability)**.** *Under linear demand with fixed markups, if* $\frac{a_1-b_1m_1}{b_1} \neq \frac{a_2-b_2m_2}{b_2}$ *and* $k = L = 0$*, then* $\Pi^D(0) > \Pi^U$*.*

Proof. Under discrimination with $k = 0$, the manufacturer solves two independent concave programs, obtaining w_i^* from (6). These coincide at a single w if and only if

$$\frac{a_1 - b_1 m_1}{b_1} = \frac{a_2 - b_2 m_2}{b_2},$$

which holds on a Lebesgue-measure-zero set of (a_i, b_i, m_i). Under the stated inequality, $w_1^* \neq w_2^*$, and the constraint $w_1 = w_2$ is binding. By the envelope theorem, the constrained maximum of a smooth objective is strictly less than the unconstrained maximum when the constraint binds. □

Remark 3. The condition in Theorem 2 is generic: when $b_1 \neq b_2$, equality of the two ratios requires a knife-edge restriction on the remaining parameters.

Since $\Pi^D(k)$ is strictly decreasing and continuous in k with $\Pi^D(0) > \Pi^U$, there exists a unique threshold $\bar{k} > 0$ satisfying

$$\Pi^D(k) > \Pi^U \iff k < \bar{k}. \tag{8}$$

Proposition 4 (Threshold and comparative statics)**.** *There exists* $\bar{k} > 0$ *such that, for sufficiently small expected legal penalty* $L \geq 0$*, the manufacturer prefers variant-based discrimination iff* $k < \bar{k}$*. Moreover:*
(a) $\bar{k}$ *is increasing in* $|b_1 - b_2|$*;*
(b) $\bar{k}$ *is decreasing in* L*.*

Proof. Differentiating (7): $\partial\Pi^D/\partial k = -\frac{1}{2}\sum_i(a_i - b_i m_i - b_i(c + k)) < 0$ for $k < \min_i \hat{k}_i$. Continuity, strict monotonicity, and $\Pi^D(0) > \Pi^U$ yield a unique $\bar{k} \in (0, \min_i \hat{k}_i)$ with $\Pi^D(\bar{k}) = \Pi^U$.

(a) $\Pi^D(0) - \Pi^U$ increases in $|b_1 - b_2|$, since greater heterogeneity makes the uniform-price compromise costlier, raising $\bar{k}$.

(b) With legal risk, the break-even condition becomes $\Pi^D(\bar{k}) = \Pi^U + L$. Since Π^D is decreasing, increasing L tightens the threshold. □

Closed form for $\bar{k}$. Define $S_i(k) := a_i - b_i m_i - b_i(c + k)$, $\Sigma := \sum_i b_i$, and $R := \sum_i(a_i - b_i m_i) - c\Sigma$. Then $\bar{k}$ solves the quadratic

$$\frac{\Sigma}{4}\bar{k}^2 - \frac{1}{2}\Big(\sum_i S_i(0)\Big)\bar{k} + \Big[\sum_i \frac{S_i(0)^2}{4b_i} - \frac{R^2}{4\Sigma} - L\Big] = 0. \tag{9}$$

The economically relevant root is the smaller positive solution.

2.5 Ramsey pricing at the optimum

At the discriminatory equilibrium, the price elasticity in segment i is $\varepsilon_i = -b_i\, p_i^D/Q_i^D$, and the optimal pricing rule satisfies

$$\frac{p_i^D - (c + k)}{p_i^D} = \frac{1}{|\varepsilon_i|}, \tag{10}$$

the standard Ramsey inverse-elasticity condition [2].

3 Welfare

Under linear demand, consumer surplus in segment i at price p_i is $CS_i = Q_i^2/(2b_i)$. Total welfare under each regime:

$$W^U = \sum_i \Big[\frac{(Q_i^U)^2}{2b_i} + (p_i^U - c)\, Q_i^U\Big], \tag{11}$$

$$W^D = \sum_i \Big[\frac{(Q_i^D)^2}{2b_i} + (p_i^D - c - k)\, Q_i^D\Big]. \tag{12}$$

The standard output condition [3] extends to this setting:

$$W^D \geq W^U \iff \sum_i Q_i^D \geq \sum_i Q_i^U. \tag{13}$$

Substituting:

$$\sum_i Q_i^D - \sum_i Q_i^U = \sum_i b_i(w^* - w_i^*). \tag{14}$$

The sign depends on parameter values; discrimination does not unambiguously raise or lower welfare.

Remark 5 (Welfare cost of differentiation). When $k > 0$, the real resource cost $k \sum_i Q_i^D$ must be subtracted from W^D. The output condition (13) then becomes necessary but not sufficient; the output expansion must dominate the differentiation cost.

4 Empirical Evidence

4.1 Data

We use the Dominick's Finer Foods scanner dataset [16], a panel of store-week-UPC retail transactions from a Chicago-area supermarket chain. The laundry-detergent category yields $n = 3{,}240{,}409$ observations spanning 566 UPCs, 93 stores, and 396 weeks after standard cleaning.[1] Brands are extracted via string matching; a binary sale indicator flags promotional weeks.

4.2 Specification

The dependent variable is per-unit price in cents per ounce:

$$\text{ppu}_{ijt} = \frac{\text{Price}_{ijt}/\text{QTY}_{ijt}}{\text{Size}_i} \times 100. \tag{15}$$

We estimate

$$\text{ppu}_{ijt} = \alpha + \beta\, \ln(\text{Size}_i) + \gamma_j + \delta_b + \phi\, \mathbf{1}[\text{Sale}_{ijt}] + \varepsilon_{ijt}, \tag{16}$$

where γ_j and δ_b are store and brand fixed effects. Standard errors are clustered by UPC (566 clusters).

4.3 Results

Table 1 reports four specifications.
Key findings:

[1] Observations with zero price, zero movement, or quality-flag failures are dropped. Per-unit prices in the top/bottom 0.1% are trimmed. Package sizes are converted to fluid ounces; non-convertible units (loads, count) excluded.

Table 1: Per-Unit Price Regressions: Dominick's Laundry Detergent

	(1)	(2)	(3)	(4)
ln(Size)	-3.003^{***}	-3.007^{***}	-2.820^{***}	-2.800^{***}
	(0.369)	(0.369)	(0.366)	(0.367)
Sale indicator				-0.606^{***}
				(0.079)
Constant	20.069^{***}	20.405^{***}	17.582^{***}	17.613^{***}
	(1.656)	(1.663)	(1.623)	(1.624)
Store FE	No	Yes	Yes	Yes
Brand FE	No	No	Yes	Yes
R^2	0.382	0.385	0.470	0.474
Adj. R^2	0.382	0.385	0.470	0.474
N	3,240,409	3,240,409	3,240,409	3,240,409

Notes: Dependent variable is per-unit price in cents per ounce. Standard errors clustered by UPC in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Store FE: 93 store dummies. Brand FE: 23 brand dummies.

(i) **Size gradient.** Baseline: $\hat{\beta} = -3.003$ (SE$= 0.369$, $p < 0.001$), $R^2 = 0.382$. A one-log-unit increase in size (roughly $\times 2.7$) is associated with a 3.0 ¢/oz decrease in per-unit price.

(ii) **Store heterogeneity.** Joint F-test on 92 store dummies: $F = 15.04$ ($p < 0.001$). Cross-store variation persists after controlling for size and brand. Dominick's operated 16 pricing zones collapsing into four tiers [17]; store FE capture a mixture of zone policy and local demand.

(iii) **Brand effects.** Adding brand FE raises R^2 from 0.385 to 0.470 ($\Delta R^2 \approx 0.09$); $\hat{\beta}$ attenuates to -2.820. The residual within-brand size gradient remains large and significant.

(iv) **Sale control.** The sale indicator enters at -0.606 (SE$= 0.079$). $\hat{\beta}$ is robust at -2.800.

Robustness. Table 2 re-estimates the full specification under store-level and two-way (UPC×store) clustering [10]. Point estimates are unchanged. Under two-way clustering, standard errors are close to the UPC-only baseline, indicating that UPC-level dependence dominates.

Table 2: Robustness Checks: Alternative Clustering Schemes (Full Model)

	(1) UPC Clustered	(2) Store Clustered	(3) Two-Way Clustered (UPC × Store)
ln(Size)	-2.800***	-2.800***	-2.800***
	(0.367)	(0.012)	(0.365)
Sale indicator	-0.606***	-0.606***	-0.606***
	(0.079)	(0.015)	(0.080)
Store FE	✓	✓	✓
Brand FE	✓	✓	✓
R^2	0.474	0.474	0.474
N		3,240,409	

Notes: Dependent variable: per-unit price (cents/oz). All three columns use the same OLS regression with store and brand fixed effects; only the covariance estimator differs. Cluster-robust standard errors in parentheses. Column (3) applies the Cameron, Gelbach, and Miller (2011) two-way formula: $\hat{V}_{\text{two-way}} = \hat{V}_{\text{UPC}} + \hat{V}_{\text{Store}} - \hat{V}_{\text{UPC}\times\text{Store}}$. Point estimates are identical across columns. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Within-brand regressions. Table 3 restricts the sample to Tide, All, and Purex separately. The size gradient persists within each brand ($\hat{\beta}_{\text{Tide}} = -0.74$, $\hat{\beta}_{\text{All}} = -1.22$, $\hat{\beta}_{\text{Purex}} = -3.13$; all $p < 0.01$), ruling out across-brand composition as the sole driver.

Table 3: Within-Brand Size–Price Gradient

	Tide	All	Purex
ln(Size)	-0.740^{***}	-1.223^{***}	-3.129^{***}
	(0.178)	(0.401)	(0.817)
Sale indicator	-0.808^{***}	-0.186	-0.425^{*}
	(0.077)	(0.130)	(0.255)
Store FE	Yes	Yes	Yes
R^2	0.231	0.209	0.492
Adj. R^2	0.230	0.209	0.492
N	564,067	237,357	104,329
UPCs	78	51	22

Notes: Dependent variable is per-unit price in cents per ounce. Each column restricts the sample to a single brand. Standard errors clustered by UPC in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. All specifications include store fixed effects and a sale indicator.

Placebo. Table 4 regresses both per-unit price and total shelf price on ln(Size). The coefficient is strongly negative for per-unit price (-2.80) and strongly positive for shelf price ($+3.89$), confirming a quantity-discount schedule rather than a mechanical artefact.

4.4 Limitations

The Dominick's data cover a single retail chain; we observe within-chain variation rather than the cross-chain discrimination central to Robinson–Patman. Wholesale prices are unobserved. Package size is endogenous. Ounce-based normalisation may not equalize washing capacity across formulations; load-equivalent pricing would be more rigorous but

Table 4: Placebo Check: Per-Unit Price vs. Total Shelf Price

	(1) PPU (¢/oz)	(2) Shelf price ($)
ln(Size)	-2.800^{***}	3.893^{***}
	(0.367)	(0.303)
Sale indicator	-0.606^{***}	-0.662^{***}
	(0.079)	(0.088)
Store FE	Yes	Yes
Brand FE	Yes	Yes
R^2	0.474	0.652
Adj. R^2	0.474	0.652
N	3,240,409	3,240,409

Notes: Column (1) uses per-unit price (cents per ounce) as the dependent variable. Column (2) uses total shelf price (dollars). The negative coefficient in column (1) and the positive coefficient in column (2) confirm that larger packages have lower per-unit prices but higher total prices, ruling out a mechanical coding artefact. Standard errors clustered by UPC in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

load counts are unavailable. The Dominick's data do not observe wholesale contracts, retailer-specific manufacturer variants, or cross-chain product allocation. Accordingly, the evidence documents retail pricing patterns *consistent with* the model's mechanism but does not directly test the wholesale discrimination channel. Causal identification would require an instrument for package-size choices or matched wholesale–retail data spanning multiple chains.

5 Discussion

The model and data are motivated by the institutional setting of the Robinson–Patman Act, but the formal structure is more general: any common-price constraint that a firm can relax via low-cost product differentiation produces the same trade-off between differentiation cost k, profit gain, and legal risk L.

Relevant applications include:

- *Detergent packaging.* Manufacturers offer sizes from 50 oz to 400 oz; the Dominick's data show per-unit prices ranging from 5 to 12 ¢/oz within Tide alone.
- *Branded vs. authorized generics.* Chemically identical pharmaceuticals sold under distinct trade dress correspond to variants A, B with different demand parameters.
- *Retailer-exclusive SKUs.* Mattress models with cosmetic differences (quilting, fabric covers) create retailer-specific UPCs, raising search costs and sustaining price dispersion.
- *International editions.* Textbook publishers sell downgraded editions in elastic foreign markets, using physical differences to segment and limit arbitrage.

From a welfare perspective, variant-based discrimination can expand output in elastic segments but sustains higher prices in inelastic ones. The net effect depends on whether total output rises [3, 2]. When $k > 0$, the social cost of differentiation further erodes the potential welfare gain (Theorem 5). Enforcement narrowly targeting "identical commodities" may therefore induce *more* cosmetic differentiation rather than less discrimination.

6 Conclusion

We model how product modification enables a manufacturer to circumvent a common-wholesale-price constraint and implement third-degree price discrimination. Closed-form

results characterize the profit gain, threshold cost $\bar{k}$, and Ramsey pricing at the optimum. Scanner data covering 3.24 M laundry-detergent transactions yield a steep size–price gradient ($\hat{\beta} = -3.0$), significant store heterogeneity ($F = 15.04$), and persistent within-brand gradients. Extensions to endogenous retailer pricing, multi-chain data, and instrumental-variable identification are left for future work.

A Asymmetric markups

With $m_1 \neq m_2$, the uniform-price optimum becomes $w^* = \frac{(a_1+a_2)-(b_1m_1+b_2m_2)+c(b_1+b_2)}{2(b_1+b_2)}$ and the discriminatory optima remain $w_i^* = \frac{a_i-b_im_i+b_i(c+k)}{2b_i}$. The profit gain $\Pi^D(0) - \Pi^U > 0$ whenever $w_1^* \neq w_2^*$, which is generically satisfied when $b_1 \neq b_2$ or $m_1 \neq m_2$. All comparative statics carry through.

B Simulation figures

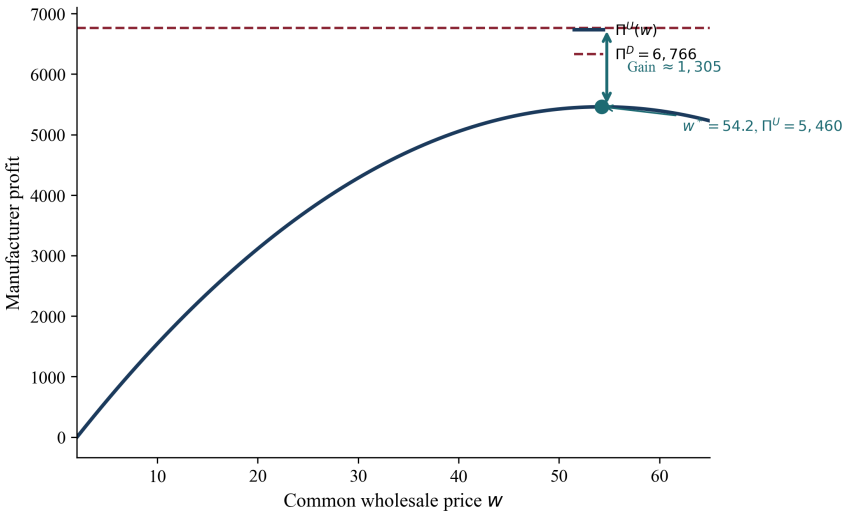

Figure 1: Manufacturer profit $\Pi^U(w)$ under uniform pricing. Dashed line: Π^D under variant-based discrimination.

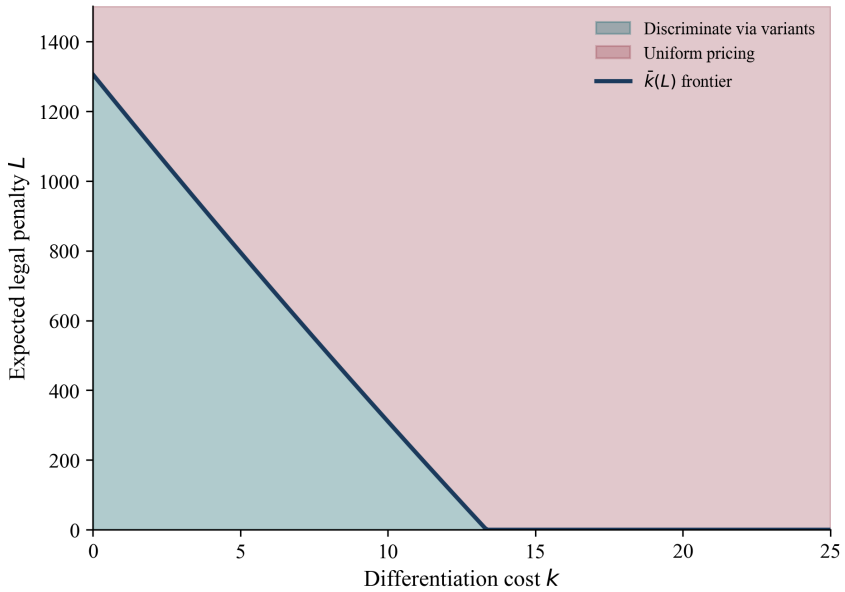

Figure 2: Strategy frontier in (k, L)-space (Theorem 4). Blue region: discrimination optimal. Red region: uniform pricing dominates.

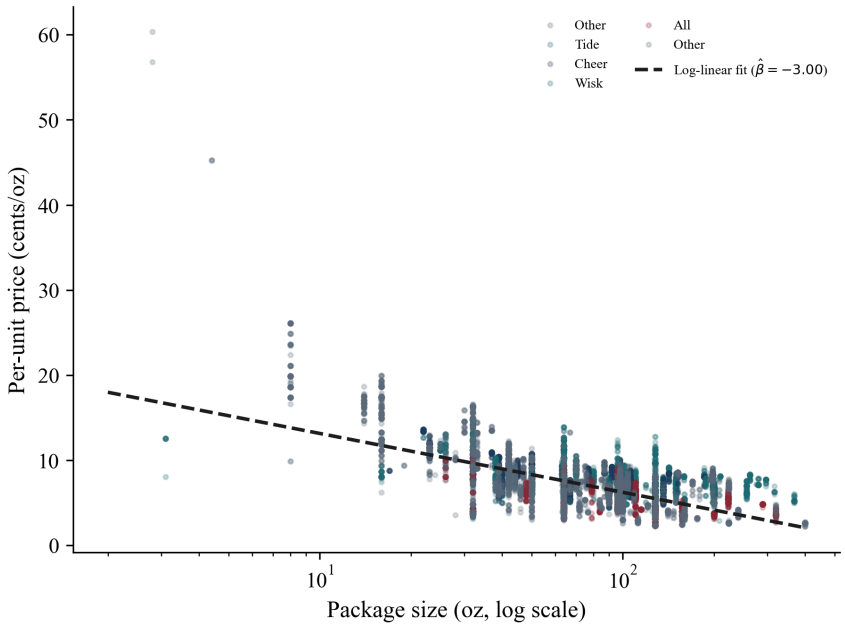

Figure 3: Per-unit price vs. package size (log scale). Points colored by brand; dashed line: log-linear fit.

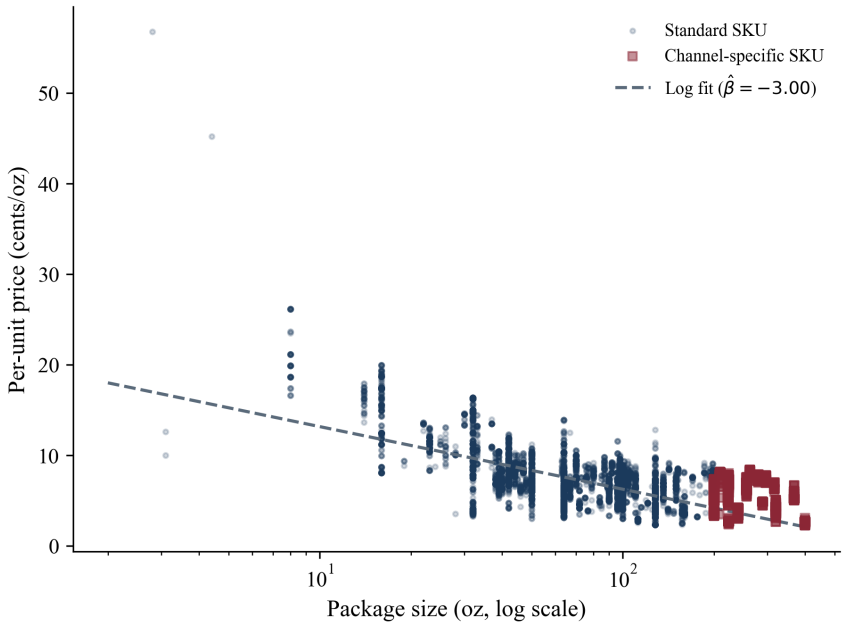

Figure 4: Per-unit price vs. size: standard SKUs (circles) and top-5% large-format SKUs (squares).

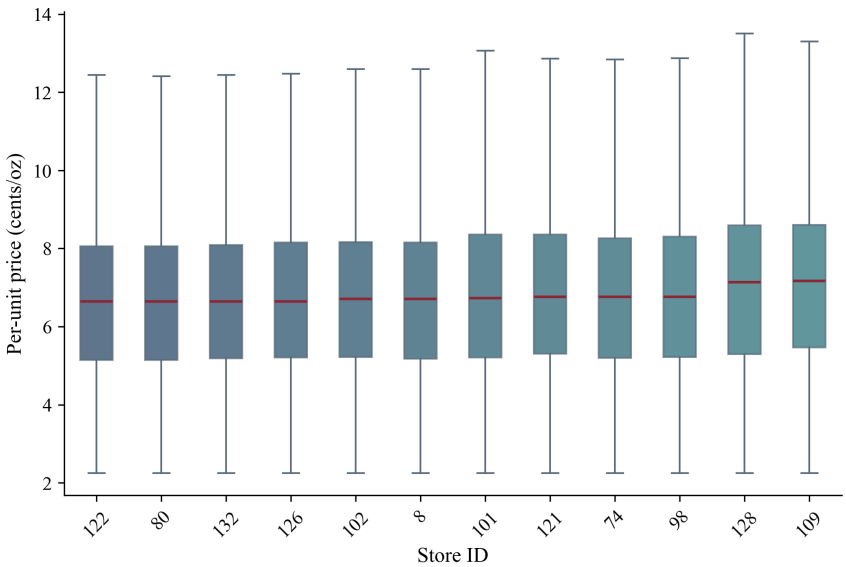

Figure 5: Per-unit price by store (12 highest-volume stores). Outliers suppressed.

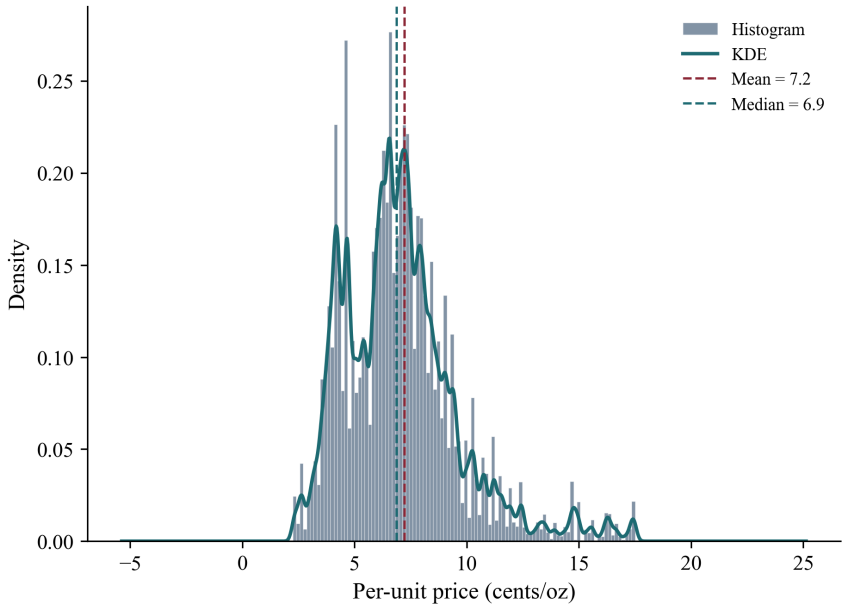

Figure 6: Distribution of per-unit prices. Vertical lines: mean and median.

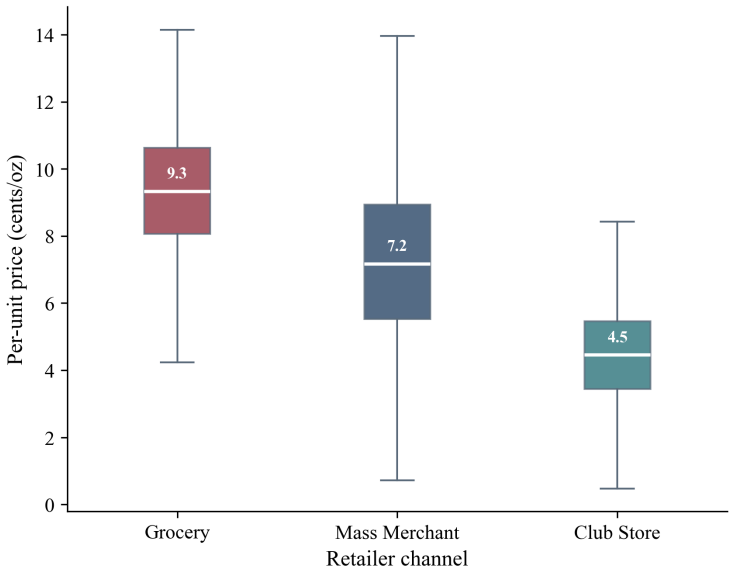

Figure 7: Simulated cross-channel price dispersion, calibrated to Dominick's data and documented retail margins.

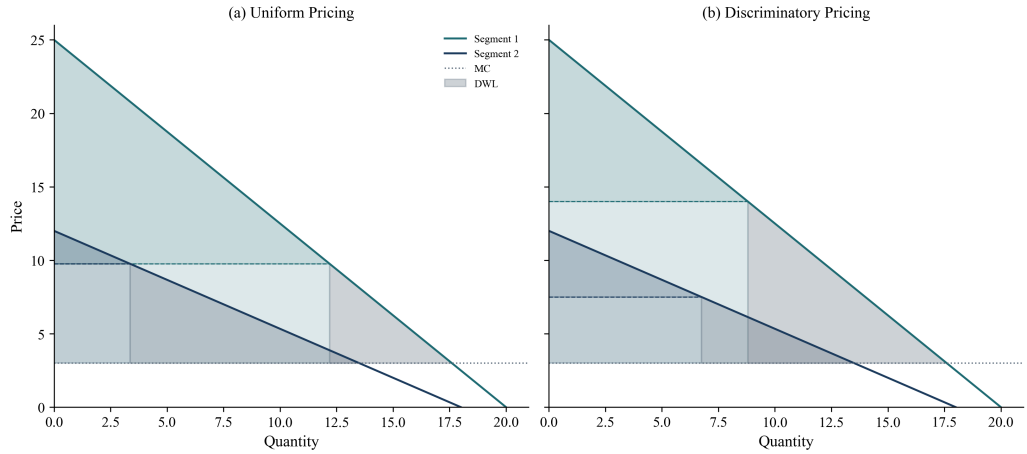

Figure 8: Welfare decomposition: (a) uniform pricing; (b) discriminatory pricing. Grey: deadweight loss.

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Data availability

The Dominick's Finer Foods dataset is publicly available at https://www.chicagobooth.edu/research/kilts/research-data/dominicks. Replication code: https://github.com/n33levo/PMandPD.

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